

AI Tool Prompt Engineering Report



Ariño, Cedric D.

BS Computer Science - 2B

21st Century IT Skills

AI Tool Overview

The AI tool I used for this activity is Leonardo.Ai, which falls under the image synthesis domain. This type of AI focuses on generating images based on text prompts.

I chose this tool because its outputs are visual, making it easier to directly observe how changes in prompt design affect the results. Compared to text-based AI tools, image generators make the differences between prompts more obvious and easier to analyze.

Original Prompt

“create a glowing tesseract suspended and rotating in space. it should appear as though it were offering a glimpse into another dimension or reality, with light bending and leaking from within it. it should feel like a visualization of the concept of infinity”



AI-Assisted Prompt

LLM Used: ChatGPT

“Create a highly detailed, glowing tesseract (4D hypercube) suspended in deep space, slowly rotating and unfolding as if revealing higher-dimensional geometry. The structure should consist of multiple nested, semi-transparent cubes connected by luminous edges, with impossible, non-Euclidean perspectives that suggest four-dimensional space. Inside the tesseract, depict an infinite recursive space, like fractal-like patterns, repeating geometric structures, or endless corridors that extend beyond visible limits, giving the illusion of boundless depth. Light should emanate from within the core, bending and refracting outward through the structure, creating soft volumetric rays, glowing particles, and subtle lens distortions. The light should feel alive, as if energy from another dimension is leaking into reality. Surround the tesseract with a dark cosmic void or minimal starfield to emphasize contrast and isolation. Use a cinematic, high-contrast lighting style with neon or ethereal hues (blues, purples, and whites), and a sense of scale that makes the object feel both infinite and incomprehensible. The overall mood should be surreal, mysterious, and awe-inspiring, like a visualization of infinity and higher-dimensional existence.”



Comparison and Analysis

After comparing the two outputs, I noticed several key differences in terms of visual style, clarity, and overall impact. The image generated from the original prompt appears more visually striking and aesthetically appealing. It has a stronger sense of brightness and depth, and the glowing effect of the tesseract feels more intense and engaging. On the other hand, the image generated from the AI-assisted prompt is more structured and controlled, but it appears less dynamic and somewhat flatter in comparison.

In terms of usefulness and accuracy, the second output aligns more closely with the technical description provided in the improved prompt. It includes clearer geometric structure, more defined edges, and a more consistent interpretation of a tesseract as a nested or layered cube. However, despite being more accurate to the prompt, it feels less impactful visually. The first image, although based on a simpler prompt, better captures the abstract idea of infinity through its brightness, depth, and overall composition.

The improved prompt significantly increased clarity and depth. It gave specific details which likely guided the AI to produce a more controlled and technically consistent image. However, this added level of detail may have also limited the model's creative freedom, resulting in a less visually compelling output. In contrast, the original prompt was more open-ended, allowing the AI to interpret the concept more freely, which may have contributed to a more visually interesting result.

One important observation I've made is that adding more details does not necessarily guarantee a better outcome. While the refined prompt improved structure and relevance to the concept of higher-dimensional geometry, it may have overconstrained the generation process. I also observed that reusing similar prompts or generating multiple iterations can sometimes lead to reduced image quality or less variation, which may have influenced the second result.

From this activity, I learned that effective prompt engineering is not just about adding more detail, but about balancing specificity and creative freedom. A well-designed prompt should guide the AI without overly restricting it. In some cases, simpler prompts can produce more aesthetically pleasing results, especially for abstract concepts where interpretation plays a major role.